

Raghav Sejpai

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LinkedIn | Github | LeetCode



Education

Vellore Institute of Technology Vellore, Tamil Nadu
Bachelor of Technology, Computer Science and Engineering (AI & ML) Aug 2023 – Present

Work Experience

Software Engineering Intern Aug 2025
Inwesol Remote

- Developed and deployed **17+ scalable REST APIs** using **Next.js** and **PostgreSQL (NeonDB)** with **99.5% uptime**
- Optimized **Drizzle ORM** queries and access patterns, improving **backend response time by 30%**
- Applied caching and concurrency-handling strategies, reducing **server load by 40%**
- Collaborated via Git-based workflows including code reviews, branching strategies, and issue tracking

Technologies: Next.js, PostgreSQL

Projects

YieldAI: MLCC Copilot | PyTorch, ONNX, TypeScript, Next.js, Zod 🔗🔗
Physics-Informed AI & Quality Control

- Trained **Physics-Informed Neural Networks (PINNs)** on 50K samples, replacing FEA simulations for **Multi-Layer Ceramic Capacitors (MLCC)** with sub-millisecond inference.
- Deployed a custom **CNN** for automated visual inspection, achieving **96% precision** and **94% recall** on defects.
- Engineered an **Adam Optimizer** for inverse-design auto-tuning; served **ONNX** models in Node.js with **Zod** validation.

Fake Document Detection System | Python, TensorFlow, OpenCV, Grad-CAM 🔗
AI-Powered Academic Document Authenticator

- Implemented an **end-to-end document forgery detection pipeline** in Python classifying academic documents as **GENUINE** or **FAKE** with **92% accuracy** using CNN and Random Forest ensembling.
- Implemented **Explainable AI (Grad-CAM)** heatmaps to localize tampered regions, achieving about **0.91 F1 score**.

textorigin | Python, RoBERTa, T5, FastAPI 🔗
AI Text Origin Classifier

- Classified human, AI, and paraphrased text via 6 features (perplexity, burstiness, entropy, etc).
- Fine-tuned **RoBERTa-base** on **15,000 samples**, synthesizing 5,000+ samples, removing hand-labelling dependency.
- Deployed **FastAPI** REST API and CLI with text, file, and JSON input modes for flexible integration.

Society and Organizational Role

Chairperson Nov 2025 – Present
IEEE Computer Society – VIT Vellore, Tamil Nadu

- User Acquisition:** Spearheaded end-to-end user acquisition by driving highly targeted top-of-funnel marketing campaigns across multiple channels, successfully onboarding **1500+ active participants** across 5 events.
- B2B Partnerships:** Drove B2B partnerships and revenue generation through a high-conversion outbound sales strategy targeting **1,400+ companies**, successfully securing **Rs. 3,95,000** in sponsorships alongside industry guest talent.
- Cross-Functional Leadership:** Exercised cross-functional leadership by managing a **250-member** organization, meticulously overseeing end-to-end event logistics, promotional marketing, and technical project delivery spanning AI and ML.

Organizing Committee Member Vellore, Tamil Nadu
Riviera – Cultural Fest, VIT

- Managed logistics and directed technical operations, delivering a seamless concert experience for **8,000+ attendees**

Organizing Committee Member Vellore, Tamil Nadu
Gravitas – Technical Fest, VIT

- Drove event registrations and optimized end-to-end logistics to reduce operational friction for attendees

Skills and Competencies

Languages & Frameworks	C++, Python, TypeScript, JavaScript, Java, Next.js, Node.js, Express.js
Databases	PostgreSQL, MySQL, MongoDB
Tools	Git, ONNX Runtime, VS Code, CI/CD basics, AWS fundamentals
AI/ML	scikit-learn, TensorFlow, OpenCV, NLP, Computer Vision, Prompt Engineering, RAG
Competitive Programming	460+ problems solved on <u>LeetCode</u> (C++)